Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Unphysicality of Hilbert spaces for quantum mechanics

Files.

- File: `briefs/HilbertUnphysicality/HilbertUnphysicality.lean`
- Library: `HilbertUnphysicality`
- Imports: `InconsistentConvergence` and three new `ForMathlib` files:
  `ForMathlib/Analysis/SpecialFunctions/Gaussian/Erf.lean` (the error function),
  `ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (separable Hilbert spaces are $\ell^2(\mathbb{N})$),
  `ForMathlib/MeasureTheory/Function/L2Space.lean` (L^2 is infinite-dimensional)

Setting. Theorem 1 says that the Hilbert-space formulation QSTO:H is inconsistent with three physical conditions. The conditions are informal; what is checked is the mathematics behind each of the three arguments.

Results.

Brief	Lean
[B]COBS	
unbounded ⇒ not continuous	`not_continuous_of_unbounded`
oscillator example	`numberExp_not_continuousAt` (from the inconsistent-convergence brief)
[B]WDQ	
Hellinger–Toeplitz	`bounded_of_isSymmetric`, `not_isSymmetric_of_unbounded`
erf, $\lvert\operatorname{erf}\rvert < 1$	`Real.erf`, `Real.abs_erf_lt_one`, `Real.tendsto_erf_atTop`
$y(x) = \tan(\frac{\pi}{2}\operatorname{erf} x)$ is a change of coordinates	`frameChange`, `strictMono_frameChange`
eq. (2), first line, integrated	`arctan_frameChange`: $\arctan(y)/\pi = \operatorname{erf}(x)/2$
eq. (2), second line	`cauchyDensity_frameChange_mul_deriv`: $\lvert\phi(y(x))\rvert^2\, y'(x) = \lvert\psi(x)\rvert^2$
ψ normalized, in the domain of X	`integral_gaussDensity`, `integrable_sq_mul_gaussDensity`
ϕ normalized, not in the domain of Y	`integral_cauchyDensity`, `not_integrable_sq_mul_cauchyDensity`
[B]STDIST	
separable Hilbert spaces are isomorphic	`nonempty_linearIsometryEquiv_of_separable`
$L^2(\mathbb{R}^n)$ is separable and infinite-dimensional	`L2.separableSpace_euclidean`, `L2.not_finiteDimensional_euclidean`
$L^2(\mathbb{R}^{n+1}) \cong L^2(\mathbb{R}^{m+1})$	`L2_euclidean_iso`

Differences from the brief.

- The wave functions are handled through their densities $|\psi|^2 = e^{-x^2}/\sqrt{\pi}$ and $|\phi|^2 = 1/(\pi(1+y^2))$. "In the domain of the position operator" is read as $\int x^2|\psi|^2 < \infty$, i.e. $x\psi \in L^2$.
- The frame change is checked as a strictly increasing coordinate change satisfying eq. (2) in both forms. The unitary U between the two L^2 spaces and the change of variables in the integral are not formalized.
- Mathlib has no error function; `Real.erf` is defined in `ForMathlib`.
- [B]STDIST is proved for $n, m \geq 1$ through the classification of separable infinite-dimensional Hilbert spaces, which is also new in `ForMathlib`.
- The additional remarks of Section 4 are not formalized.

Problems found in the brief.

- The brief says that $\phi(y)$ "goes to zero at infinity like $1/y^2$". It is $|\phi|^2$ that decays like $1/y^2$; ϕ itself decays like $1/|y|$. The conclusion is unaffected: $y\,\phi(y) \notin L^2$.
- "$L^2(\mathbb{R}^n)$ is a separable Hilbert space for any n" is true, but the isomorphism argument also needs infinite dimension, which fails for $n = 0$.